

Institute/Department	UNIVERSITY INSTITUTE OF COMPUTING (UIC)	Program	Master of Computer Applications - Artificial Intelligence & Machine Learning(MC306)
Master Subject Coordinator Name:	Nisha Sharma	Master Subject Coordinator E-Code:	E2915
Course Name	Artificial Intelligence	Course Code	25CAT-613

Lecture	Tutorial	Practical	Self Study	Credit	Subject Type
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Course Type	Course Category	Mode of Assessment	Mode of Delivery
Major Core	Graded (GR)	Theory Examination (ET)	Theory (TH)

Mission of the Department	M1 To provide innovative learning centric facilities and quality-oriented teaching learning process for solving computational problems. M2 To provide a frame work through Project Based Learning to support society and industry in promoting a multidisciplinary activity. M3To develop crystal clear evaluation system and experiential learning mechanism aligned with futuristic technologies and industry. M4 To provide doorway for promoting research, innovation and entrepreneurship skills in collaboration with industry and academia. M5 To undertake societal activities for upliftment of rural/deprived sections of society
Vision of the Department	To be a Centre of Excellence for nurturing computer professionals with strong application expertise through experiential learning and research for matching the requirements of industry and society instilling in them the spirit of innovation and entrepreneurship.

Program Educational Objectives(PEOs)

PEO1	Demonstrate analytical and design skills including the ability to generate creative solutions and foster team-oriented professionalism through effective communication in their careers.
PEO2	Expertise in successful careers based on their understanding of formal and practical methods of application development using the concept of computer programming languages and design principles in accordance to industry 4.0
PEO3	Exhibit the growth of the nation and society by implementing and acquiring knowledge of upliftment of health, safety and other societal issues.
PEO4	Implement their exhibiting critical thinking and problem- solving skills in professional practices or tackle social, technical and business challenges.

Program Specific OutComes(PSOs)

PSO1	Analyze their abilities in systematic planning, developing, testing and executing complex computing applications in field of Social Media and Analytics, Web Application Development and Data Interpretations.
PSO2	Apprise in-depth expertise and sustainable learning that contributes to multi-disciplinary creativity, permutation, modernization and study to address global interest.

Program OutComes(POs)

PO1	Apply mathematics and computing fundamental and domain concepts to find out the solution of defined problems and requirements.
PO2	Use fundamental principle of Mathematics and Computing to identify, formulate research literature for solving complex problems, reaching appropriate solutions.
PO3	Understand to design, analyse and develop solutions and evaluate system components or processes to meet specific need for local, regional and global public health, societal, cultural, and environmental systems.
PO4	Use expertise research-based knowledge and methods including skills for analysis and development of information to reach valid conclusions.
PO5	Adapt, apply appropriate modern computing tools and techniques to solve computing activities keeping in view the limitations
PO6	Exhibiting ethics for regulations, responsibilities and norms in professional computing practices.

PO7	Enlighten knowledge to enhance understanding and building research, strategies in independent learning for continual development as computer applications professional.
PO8	Establishing strategies in developing and implementing ideas in multi- disciplinary environments using computing and management skills as a member or leader in a team.
PO9	Contribute to progressive community and society in comprehending computing activities by writing effective reports, designing documentation, making effective presentation, and understand instructions
PO10	Apply mathematics and computing knowledge to access and solve issues relating to health, safety, societal, environmental, legal, and cultural issues within local, regional and global context.
PO11	Gain confidence for self and continuous learning to improve knowledge and competence as a member or leader of a team.
PO12	Learn to innovate, design and develop solutions for solving real life business problems and addressing business development issues with a passion for quality competency and holistic approach.

Text Books					
Sr No	Title of the Book	Author Name	Volume/Edition	Publish Hours	Years
1	Artificial Intelligence: A Modern Approach	Stuart Russell and Peter Norvig	4th	pearson	2020
2	Artificial Intelligence: Foundations of Computational Agents	David L. Poole and Alan K. Mackworth	2nd	Cambridge University Press	2017
3	Python Artificial Intelligence Projects for Beginners	Joshua Eckroth	1st	Packt Publishing	2018

Reference Books					
Sr No	Title of the Book	Author Name	Volume/Edition	Publish Hours	Years
1	Artificial Intelligence: Structures and Strategies for Complex Problem Solving	George F. Luger	6th	Pearson	2008
2	Machine Learning: A Probabilistic Perspective	Kevin P. Murphy	1st	MIT Press	2012
3	Planning Algorithms	Steven M. LaValle	1st	Cambridge University Press	2006

Course OutCome	
SrNo	OutCome
CO1	Gain a foundational understanding of Artificial Intelligence (AI) concepts and applications
CO2	Analyze and formulate single-agent search problems in state space representation.
CO3	Apply various search algorithms to solve problems using Python.
CO4	Evaluate the efficiency and effectiveness of different search algorithms based on problem characteristics
CO5	Developing AI agents that can play games by utilizing search algorithms to make optimal moves

Lecture Plan Preview-Theory						
Unit No	LectureNo	ChapterName	Topic	Text/ Reference Books	Pedagogical Tool**	Mapped with CO Numer (s)
1	1	Introduction to AI	Definition, history	,T-Artificial Intelligence: A Mod,R-Artificial Intelligence: Struc	PPT	CO1
1	2	Introduction to AI	future of AI	,T-Artificial Intelligence: A Mod,R-Artificial Intelligence: Struc	PPT	CO1
1	3	applications of search in AI	Applications of various types of search techniques in AI	,T-Artificial Intelligence: A Mod,R-Artificial Intelligence: Struc	PPT,Video Lecture	CO1
1	4	Intelligent Agents	Characteristics of Intelligent Agents	,T-Artificial Intelligence: A Mod,R-Artificial Intelligence: Struc	PPT	CO1

1	5	Intelligent Agents	Characteristics of Intelligent Agents	,T-Artificial Intelligence: A Mod,R- Artificial Intelligence: Struc	Case Study,PPT,V ideo Lecture	CO1
1	6	Intelligent Agents	Rationality	,T-Artificial Intelligence: A Mod,T- Artificial Intelligence: Found,R- Artificial Intelligence: Struc,R- Machine Learning: A Probabilis	PPT,Simulati on,Video Lecture	CO1
1	7	Intelligent Agents	problem-solving abilities	,T-Artificial Intelligence: A Mod,T- Artificial Intelligence: Found,R- Artificial Intelligence: Struc,R- Machine Learning: A Probabilis	Case Study,PPT,R eports,Video Lecture	CO1
1	8	Single-Agent Search Problems	State space representation	,T-Artificial Intelligence: A Mod,T- Artificial Intelligence: Found,R- Artificial Intelligence: Struc,R- Machine Learning: A Probabilis	Activity,Case Study,PPT,V ideo Lecture	CO2
1	9	Single-Agent Search Problems	goal state	,T-Artificial Intelligence: A Mod,T- Artificial Intelligence: Found,T- Python Artificial Intelligence,R- Artificial Intelligence: Struc,R- Machine Learning: A Probabilis,R- Planning Algorithms	Case Study,PPT,V ideo Lecture	CO2
1	10	Single-Agent Search Problems	problem formulation	,T-Artificial Intelligence: A Mod,T- Artificial Intelligence: Found,R- Artificial Intelligence: Struc,R- Machine Learning: A Probabilis	PPT,Video Lecture	CO2
1	11	Uninformed Search Techniques	Breadth-First Search (BFS),	,T-Artificial Intelligence: A Mod,T- Artificial Intelligence: Found,R- Artificial Intelligence: Struc,R- Machine Learning: A Probabilis	PPT,Video Lecture	CO3
1	12	Uninformed Search Techniques	Depth-First Search (DFS)	,T-Artificial Intelligence: A Mod,T- Artificial Intelligence: Found,R- Artificial Intelligence: Struc,R- Machine Learning: A Probabilis	Case Study,PPT,V ideo Lecture	CO3
1	13	Uninformed Search Techniques	Iterative Deepening Search (IDS).	,T-Artificial Intelligence: A Mod,T- Artificial Intelligence: Found,R- Artificial Intelligence: Struc,R- Machine Learning: A Probabilis	Case Study,PPT,V ideo Lecture	CO3
1	14	Uninformed Search Techniques	comparison of Informed Search Techniques	,T-Artificial Intelligence: A Mod,T- Artificial Intelligence: Found,T- Python Artificial Intelligence,R- Artificial Intelligence: Struc,R- Machine Learning: A Probabilis,R- Planning Algorithms	PPT,Video Lecture	CO3
1	15	Uninformed Search Techniques	uniform-cost search	,T-Artificial Intelligence: A Mod,T- Artificial Intelligence: Found,T- Python Artificial Intelligence,R- Artificial Intelligence: Struc,R- Machine Learning: A Probabilis,R- Planning Algorithms	PPT,Video Lecture	CO3
2	16	Heuristic Search Basics	Heuristic functions.	,T-Artificial Intelligence: A Mod,T- Artificial Intelligence: Found,R- Artificial Intelligence: Struc,R- Machine Learning: A Probabilis,R- Planning Algorithms	Case Study,PPT,V ideo Lecture	CO3
2	17	Informed Search Techniques	evaluation functions.	,T-Artificial Intelligence: A Mod,T- Artificial Intelligence: Found,T- Python Artificial Intelligence,R- Artificial Intelligence: Struc,R- Machine Learning: A Probabilis,R- Planning Algorithms	Case Study,PPT,V ideo Lecture	CO3
2	18	Informed Search Techniques	Best - First Search Algorithm	,T-Artificial Intelligence: A Mod,T- Artificial Intelligence: Found,T- Python Artificial Intelligence,R- Artificial Intelligence: Struc,R- Machine Learning: A Probabilis,R- Planning Algorithms	PPT,Video Lecture	CO3

2	19	Informed Search Techniques	Beam Search Algorithm	,T-Artificial Intelligence: A Mod,T-Artificial Intelligence: Found,T-Python Artificial Intelligence,R-Artificial Intelligence: Struc,R-Machine Learning: A Probabilis,R-Planning Algorithms	Case Study,PPT,V ideo Lecture	CO3
2	20	A* and AO* Algorithms	A* Search Algorithm	,T-Artificial Intelligence: A Mod,T-Artificial Intelligence: Found,T-Python Artificial Intelligence,R-Artificial Intelligence: Struc,R-Machine Learning: A Probabilis,R-Planning Algorithms	PPT,Video Lecture	CO3
2	21	A* and AO* Algorithms	Combining informed and uninformed search for efficiency	,T-Artificial Intelligence: A Mod,T-Artificial Intelligence: Found,T-Python Artificial Intelligence,R-Artificial Intelligence: Struc,R-Machine Learning: A Probabilis,R-Planning Algorithms	Video Lecture	CO4
2	22	A* and AO* Algorithms	Problem Reduction	,T-Artificial Intelligence: A Mod,T-Artificial Intelligence: Found,T-Python Artificial Intelligence,R-Artificial Intelligence: Struc,R-Machine Learning: A Probabilis,R-Planning Algorithms	PPT,Video Lecture	CO4
2	23	A* and AO* Algorithms	AO* Search Algorithm,	,T-Artificial Intelligence: A Mod,T-Artificial Intelligence: Found,T-Python Artificial Intelligence,R-Artificial Intelligence: Struc,R-Machine Learning: A Probabilis,R-Planning Algorithms	PPT,Video Lecture	CO3
2	24	Informed Search Techniques	Key Features of Heuristic Functions	,T-Artificial Intelligence: A Mod,T-Artificial Intelligence: Found,T-Python Artificial Intelligence,R-Artificial Intelligence: Struc,R-Machine Learning: A Probabilis,R-Planning Algorithms	PPT,Video Lecture	CO3
2	25	Heuristic Search Basics	Key Features of Heuristic Functions	,T-Artificial Intelligence: A Mod,T-Artificial Intelligence: Found,T-Python Artificial Intelligence,R-Artificial Intelligence: Struc,R-Machine Learning: A Probabilis,R-Planning Algorithms	PPT	CO3
2	26	Informed Search Techniques	Examples of Heuristic Functions: 8-Puzzle, Pathfinding, Traveling Salesman Problem	,T-Artificial Intelligence: A Mod,T-Artificial Intelligence: Found,T-Python Artificial Intelligence,R-Artificial Intelligence: Struc,R-Machine Learning: A Probabilis,R-Planning Algorithms	PPT,Video Lecture	CO4
2	27	Informed Search Techniques	Comparison of Search Techniques	,T-Artificial Intelligence: A Mod,T-Artificial Intelligence: Found,T-Python Artificial Intelligence,R-Artificial Intelligence: Struc,R-Machine Learning: A Probabilis,R-Planning Algorithms	PPT,Video Lecture	CO4
2	28	Informed Search Techniques	Pros and Cons of informed search techniques	,T-Artificial Intelligence: A Mod,T-Artificial Intelligence: Found,T-Python Artificial Intelligence,R-Artificial Intelligence: Struc,R-Machine Learning: A Probabilis,R-Planning Algorithms	PPT,Video Lecture	CO1
2	29	A* and AO* Algorithms	Key Differences Between A* and AO*	,T-Artificial Intelligence: A Mod,T-Artificial Intelligence: Found,T-Python Artificial Intelligence,R-Artificial Intelligence: Struc,R-Machine Learning: A Probabilis,R-Planning Algorithms	PPT,Video Lecture	CO4

2	30	A* and AO* Algorithms	Applications of A* & AO*	,T-Artificial Intelligence: A Mod,T-Artificial Intelligence: Found,T-Python Artificial Intelligence,R-Artificial Intelligence: Struc,R-Machine Learning: A Probabilis,R-Planning Algorithms	PPT,Video Lecture	CO3
3	31	Local Search	Hill Climbing Algorithm	,T-Artificial Intelligence: A Mod,T-Artificial Intelligence: Found,T-Python Artificial Intelligence,R-Artificial Intelligence: Struc,R-Machine Learning: A Probabilis,R-Planning Algorithms	PPT,Video Lecture	CO3
3	32	Local Search	Simple Hill Climbing	,T-Artificial Intelligence: A Mod,T-Artificial Intelligence: Found,T-Python Artificial Intelligence,R-Artificial Intelligence: Struc,R-Machine Learning: A Probabilis,R-Planning Algorithms	PPT,Video Lecture	CO3
3	33	Local Search	Steepest-Ascent Hill Climbing	,T-Artificial Intelligence: A Mod,T-Artificial Intelligence: Found,T-Python Artificial Intelligence,R-Artificial Intelligence: Struc,R-Machine Learning: A Probabilis,R-Planning Algorithms	Activity,Case Study,PPT,V ideo Lecture	CO3
3	34	Local Search	Stochastic Hill Climbing	,T-Artificial Intelligence: A Mod,T-Artificial Intelligence: Found,T-Python Artificial Intelligence,R-Artificial Intelligence: Struc,R-Machine Learning: A Probabilis,R-Planning Algorithms	Activity,PPT, Video Lecture	CO3
3	35	Local Search	Challenges in Hill Climbing Algorithm	,T-Artificial Intelligence: A Mod,T-Artificial Intelligence: Found,T-Python Artificial Intelligence,R-Artificial Intelligence: Struc,R-Machine Learning: A Probabilis,R-Planning Algorithms	Case Study,PPT,V ideo Lecture	CO4
3	36	Local Search	Applications of hill climbing	T-Artificial Intelligence: A Mod,T-Artificial Intelligence: Found,T-Python Artificial Intelligence	Case Study,PPT,V ideo Lecture	CO4
3	37	Local Search	Iterative Hill Climbing Algorithm Advantages, Applications of Iterative Hill Climbing Algorithm Introduction to adversarial search minimax algorithm	,T-Artificial Intelligence: A Mod,T-Artificial Intelligence: Found,T-Python Artificial Intelligence,R-Artificial Intelligence: Struc,R-Machine Learning: A Probabilis,R-Planning Algorithms	Activity,PPT, Video Lecture	CO3
3	38	Local Search	Advantages, Applications of Iterative Hill Climbing Algorithm	,T-Artificial Intelligence: A Mod,T-Artificial Intelligence: Found,T-Python Artificial Intelligence,R-Artificial Intelligence: Struc,R-Machine Learning: A Probabilis,R-Planning Algorithms	Case Study,PPT,V ideo Lecture	CO4
3	39	Adversarial Search	Introduction to adversarial search	,T-Artificial Intelligence: A Mod,T-Artificial Intelligence: Found,T-Python Artificial Intelligence,R-Artificial Intelligence: Struc,R-Machine Learning: A Probabilis,R-Planning Algorithms	Case Study,PPT,V ideo Lecture	CO3
3	40	Adversarial Search	minimax algorithm	,T-Artificial Intelligence: A Mod,T-Artificial Intelligence: Found,T-Python Artificial Intelligence,R-Artificial Intelligence: Struc,R-Machine Learning: A Probabilis,R-Planning Algorithms	Activity,Case Study,Flippe d Classes,PPT ,Video Lecture	CO3

3	41	Adversarial Search	alpha-beta pruning	,T-Artificial Intelligence: A Mod,T-Artificial Intelligence: Found,T-Python Artificial Intelligence,R-Artificial Intelligence: Struc,R-Machine Learning: A Probabilis,R-Planning Algorithms	Case Study,Video Lecture	CO3
3	42	Adversarial Search	Key Concepts of Adversarial Search	,T-Artificial Intelligence: A Mod,T-Artificial Intelligence: Found,T-Python Artificial Intelligence,R-Artificial Intelligence: Struc,R-Machine Learning: A Probabilis,R-Planning Algorithms	PPT,Video Lecture	CO1
3	43	Adversarial Search	Benefits of Alpha-Beta Pruning	,T-Artificial Intelligence: A Mod,T-Artificial Intelligence: Found,T-Python Artificial Intelligence,R-Artificial Intelligence: Struc,R-Machine Learning: A Probabilis,R-Planning Algorithms	PPT,Video Lecture	CO1
3	44	Adversarial Search	Expected Outcome vs. Optimal Play	,T-Artificial Intelligence: A Mod,T-Artificial Intelligence: Found,T-Python Artificial Intelligence,R-Artificial Intelligence: Struc,R-Machine Learning: A Probabilis,R-Planning Algorithms	PPT,Video Lecture	CO1
3	45	Adversarial Search	Expected Outcome vs. Optimal Play	,T-Artificial Intelligence: A Mod,T-Artificial Intelligence: Found,T-Python Artificial Intelligence,R-Artificial Intelligence: Struc,R-Machine Learning: A Probabilis,R-Planning Algorithms	PPT,Video Lecture	CO1

Assessment Model			
Sr No	Assessment Name	Exam Name	Max Marks
1	20EP02	External Theory	60
2	20EP02	Attendance Marks	2
3	20EP02	Mid-Semester Test-1	20
4	20EP02	Short Term Paper / Research Paper	12
5	20EP02	Mid-Semester Test-2	20
6	20EP02	Quiz	6

CO vs PO/PSO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	NA	NA	1	NA	NA	NA	NA	NA	NA	NA	2	2
CO2	3	2	3	3	2	NA	NA	NA	NA	NA	NA	NA	2	3
CO3	2	NA	3	NA	NA	NA	2	NA	NA	NA	NA	NA	3	2
CO4	3	3	NA	1	3	NA	NA	NA	NA	NA	NA	NA	3	3
CO5	3	NA	3	2	3	NA	2	NA	NA	NA	NA	NA	2	2
Target	2.8	2.33	3	2	2.25	NA	2	NA	NA	NA	NA	NA	2.4	2.4

